

Ali Abdallah

Junior Software Engineer

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Computer Science graduate with strong foundations in OOP, Data Structures, Algorithms, Design Patterns, and Databases. Passionate about Data Science, NLP, and Machine & Deep Learning.

Education

B.Sc. in Computer Science, Ain Shams University (Class of 2024).

📅 October 2020 - July 2024

📍 Cairo, Egypt

Grade: Very Good

Graduation Project: Masha'erohom - Arabic Emotion Detector (**Grade: A+**)

Technical Skills

Programming: Python, R, C++, C#, Java, SQL (MySQL, SQLite), MongoDB, Git, Bash

Core CS: OOP, Data Structures, Algorithms, Design Patterns, System Design

Data & AI: Data Analysis, Machine Learning, Deep Learning (TensorFlow, PyTorch, Scikit-learn), NLP, Data Visualization, Statistical Analysis

Selected Projects

Masha'erohom - Arabic Emotion Detector 📄

📅 August 2023 - July 2024

- Led a team to develop an Arabic emotion detection model exceeding state-of-the-art (SOTA) accuracy (92%, +7% vs. prior work).
- Developed and architected an ensemble of Bi-LSTM, Bi-GRU, and fine-tuned MARBERTv2, stacked via Random Forest after extensive hyperparameter tuning and evaluation.
- Fine-tuned pretrained Arabic word embeddings to better capture dialectal nuances, diacritics, and morphology.
- Built a Flask web app that ingests Twitter data and visualizes public emotion distributions for user-entered keywords.
- Published a **peer-reviewed paper** ➡ following formal evaluations in IJICIS.

ElectroHub - Online Electronics Shopping System 📄

📅 November 2023 - December 2023

- Developed a full-stack e-commerce platform using Java, JSP, MySQL, applying design patterns (Singleton, Factory, Strategy).

English Sign Language Recognition 📄

📅 August 2023 - September 2023

- Developed a CNN-based model using landmark hand detection, achieving 99% test accuracy.
- Integrated Random Forest for real-time gesture classification.

Data Security Algorithms 📄

📅 February 2023 - May 2023

- Implemented 15 encryption algorithms (AES, DES, Playfair, etc.) in C#.
- Designed tests for encryption/decryption validation and performance benchmarking.

Customer Segmentation 📄

📅 November 2022 - December 2022

- Performed unsupervised learning (K-Means, DBSCAN, Hierarchical) using R for market segmentation.
- Conducted EDA and visualized insights to support business strategy.